Prathmesh Garule

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Skills

Languages: Python, PySpark, C/C++, Bash, SQL

Technologies & Tools: Azure Data Factory, Databricks, Delta Lake, Data Mesh, ARM Templates, Azure OpenAI, RAG, Agentic AI, LangChain (AutoGen), REST APIs, Git, CI/CD, Monitoring, Ansible (AZF)

Cloud & Platforms: Azure, Dataverse, Linux, Streamlit, React, Firebase

Education

COEP Tech University, Pune Aug 2020 – Jun 2024
B.Tech — Computer Engineering – Data Science, NLP, DBMS, Data Structures and Algorithms **CGPA: 8.0/10**

NMV Jr. College, Pune 2020
Class XII (HSC - PCMB - 12th) — 92.46%

New English School, Andori 2018
Class X (SSC - Science - 10th) — 94.8%

Work Experience

UBS Business Solutions — Wealth Management, Bankers Desktop

Engineer (Full-time) Aug 2024 – Present

- Designed and delivered scalable ETL pipelines using Azure Data Factory, Databricks (PySpark), Delta Lake, and SQL, including incremental loads, metadata-driven orchestration, and automated data quality checks.
- Contributed to Data Mesh and data virtualization initiatives, enabling governed cross-domain access and improved interoperability.
- **AI-related contributions:**
  - Deployed an Azure OpenAI instance for sentiment analysis on client reviews, feedback, and trend insights.
  - Built a RAG-based internal knowledge-retrieval POC (Talk2Docs) for lead identification and cross-domain use cases. (Azure AI Search, text embeddings, OpenAI)
  - Secured **Top-4 (India, Innovation Category)** in the UBS Hackathon for an Agentic AI deployment automation system using React, Streamlit, Python, LangChain, and AutoGen.
- **Engineering Initiatives:**
  - Delivered a SharePoint-to-ADLS POC enabling net-new inbound data pipelines.
  - Executed Autosys infrastructure deployment: VM provisioning, agent rollout, and GitLab-based job promotion workflows.
  - Reduced tech debt by improving init scripts, Ansible AZF components, and strengthening release & deployment processes.

Intern — WMA Bankers Desktop Team Jan 2024 – Jun 2024

- Worked on data integration using Azure Data Factory, Databricks (PySpark, SQL), Python, and Ansible. Managed data ingestion in Dataverse (D365), implemented Databricks transformations, deployed solutions via ARM templates, and followed Agile delivery practices. Contributed to BCM and Disaster Recovery processes.
- Summer Intern [WMA STAAT Analytics]: Collaborated on EDW-to-ADW repointing using SQL, Azure tooling, MicroStrategy, MS SQL Server Management, and Oracle SQL Developer.

Project Work

- **Decoding Visual Context: Speech-Based VQA System**
  - Developed a multimodal Speech-based Visual Question Answering (VQA) system integrating computer vision, NLP, and speech recognition.
  - Built a classifier-driven pipeline to categorize question–image pairs into six VQA domains and dynamically select the optimal model.
  - Achieved 96.5% accuracy using a fusion classifier combining BERT text encoding and Swin Transformer image embeddings.
  - Integrated OpenAI Whisper ASR and built an interactive app supporting image upload, spoken queries, and text/audio responses.
- **MediPlus — Healthcare App:** React Native | Node.js | Firebase Designed an Android healthcare platform with appointment scheduling, medical encyclopedia, document vault, and disease prediction using Python-based models and web-scraped datasets.
- **XV6 — EXT2 File System (CLI Editor):** C | Ncurses Developed a terminal-based text editor supporting cursor navigation, buffer handling, basic commands, and ncurses-based UI rendering.

Certificates

- Software Engineering Virtual Internship — JP Morgan Chase & Co.
- Python and Data Analysis — Shape AI
- Hacktoberfest — DigitalOcean
- Smart Pune Health Hackathon — Oasys / Pune Smart City

Position of Responsibility

- Coordinator — Robotic Study Circle (microcontroller programming, line-following bots, PCB design, Proteus simulation)
- Member — Computer Society of India (CSI)
- Publicity Coordinator — Pune Startup Festival